

Robotic Gripper Design

Project Report

Team Members

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1. Introduction

This project focuses on the design and development of a robotic gripper using Autodesk Fusion 360. The gripper is intended to perform tasks such as gripping, lifting, and releasing objects with precision and efficiency. The design emphasizes functionality, ease of integration with robotic systems, and the selection of suitable materials.

2. Objectives

- To create a functional robotic gripper using CAD modeling in Fusion 360.
- To ensure the gripper is capable of performing key tasks: gripping, lifting, and releasing.
- To facilitate seamless integration with automation systems.
- To select lightweight and durable materials suitable for fabrication and use.

3. Key Elements of the Design Process

1. **CAD Modelling (Fusion 360):** 3D design of gripper components.
2. **CAM (Computer-Aided Manufacturing):** Creating toolpaths for CNC machining.
3. **CAPP (Computer-Aided Process Planning):** Step-by-step workflow to transition from design to production.
4. **ERP (Enterprise Resource Planning):** Monitoring gripper performance and integration into larger systems.
5. **Automation:** Deployment of the gripper for practical, automated tasks.

4. CAD Model Overview

The gripper design includes:

- Configured joints and gear mechanisms.
- Mounting features compatible with robotic arms or servos.
- Functional considerations such as actuator placement and gripping range.
- Detailed tolerances, dimensions, and part geometry.

5. CIM Workflow Integration

The gripper design is incorporated into a broader Computer Integrated Manufacturing (CIM) process:

- CAD model → CAM software → CNC machining → Final assembly.
- Ensures precision, consistency, and compatibility with robotic systems.

6. Machining & Manufacturing Processes

Components and Operations:

- **Base Plate:** Facing, milling, drilling.
- **Spur Gears:** Facing, gear cutting (hobbing/milling), drilling.
- **Connecting Links:** Facing, milling, drilling.
- **Gripper Fingers:** Facing, milling, drilling, optional slotting.
- **Connecting Pin:** Turning, facing, grooving.
- **Hinges:** Facing, milling, drilling, possible bending.

7. Technical Specifications

- **Gripper Type:** Two-finger mechanical.
- **Actuation:** Single SG90 Servo Motor.
- **Mechanism:** Gear-driven linkage.
- **Material:** ABS, PLA, Aluminum (for lightweight and durability).
- **Fabrication:** 3D printing and CNC machining.
- **Simulation:** Validated using Fusion 360 motion studies.

8. CAPP - Process Planning

- **Requirement Analysis**
- **Conceptual Design**
- **Material Selection**
- **CAD Modeling & Simulation**
- **Motor Selection**
- **Fabrication**
- **Testing & Inspection**

Each step receives an equal weight of 14.3% in the overall project process.

9. Automation & ERP Integration

- The gripper is attached to a robotic arm for tasks like pick-and-place, assembly, or inspection.
- Real-time data (errors, cycle time, part count) is sent to an ERP system.
- Enables workflow optimization and efficient resource utilization.

10. Future Improvements

- **AI-Driven Gripping:** For adaptive, intelligent handling.
- **Force Sensors:** To dynamically adjust grip strength.
- **Energy Optimization:** Using efficient actuators.
- **Miniaturization:** For use in compact robotic systems.

11. Conclusion

The project successfully developed a high-performance robotic gripper that is:

- **Functionally Effective** – With validated simulations and precise control.
- **Durable and Lightweight** – Using optimal materials.
- **Scalable and Modular** – Suitable for future integration with intelligent systems.

The gripper stands as a reliable component for various robotic applications and offers room for technological enhancement.